

THIN LENSES

Optical lenses are now a part of our everyday lives. For people who are short-sighted or long-sighted, lenses enable them to see sharp images of the world. They are also widely used in cameras, projectors, and more. In this lab, you will investigate the basic relationships for a thin lens and measure its focal length.

MEASUREMENTS

SETUP:

REFRACTION ON RECTANGULAR PRISM

Find a configuration of the lens and screen that produces the sharpest image possible. Record the positions of the object x_O (metal plate), the lens x_L , and the image x_I (screen) on the scale. Repeat the measurements for at least twelve different arrangements.

x_O (cm)													$\pm$
x_L (cm)													$\pm$
x_I (cm)													$\pm$

ANALYSIS

- Graph the image distance d_I vs. the object distance d_O and fit the expected function to the data.
- Graph the reciprocal values of the image distance d_I vs. the reciprocal values of the object distance d_O and fit the expected function to the data..
- Graph the product of image and object distance vs. their sum and fit the expected function to the data.
- For each of the fit functions, determine the focal length from the fit parameters and discuss if they are compatible with each other and the nominal focal length of the lens.

TWO LENSES

Find an arrangement with two lenses where the image is upright. Sketch the setup and take notes of the relevant quantities (positions, sizes, focal lengths).

ANALYSIS

- Predict the position and the size of the observed image.

ADDITIONAL TASKS

1. Find a configuration with the two lenses used in the lab session that creates an upright image with a lateral magnification of 2.
2. Find a configuration with the two lenses used in the lab session that creates an upside-down image.
3. Making a reasonable assumption for the refractive index of the lens used in the first experiment of the lab session, calculate the radius of curvature and the thickness of the lens.
4. Using Snell's law, show that parallel rays entering a spherical lens (parallel to the optical axis) do not converge at a perfect focal point within the lens.
5. Write a computer program to calculate the paths of several light rays (based on Snell's law) to explore the quality of the focal points of different types of lenses (e.g. biconvex or plano-convex).